

Recap.

① For RVs X and Y , $E(Y|X=x) = \begin{cases} \sum_y y \cdot \frac{P_{X,Y}(x,y)}{P_X(x)} \\ \int_{-\infty}^{\infty} y \cdot \frac{P_{X,Y}(x,y)}{P_X(x)} dy \end{cases}$

② For RVs X and Y , $X \perp Y$, pdfs of X and Y are known, define $Z = X + Y$.

$$P_Z(z) = \begin{cases} \sum_y P_X(z-y) P_Y(y) \\ \int_{-\infty}^{\infty} P_X(z-y) P_Y(y) dy \end{cases}$$

③ For RV X with known cdf $F_X(x)$, to simulate it's distribution, define uniformly on $(0,1)$ distributed RV U , then $X = F_X^{-1}(U)$.

e.g. For $X \sim \text{Exp}(\lambda)$, $F_X(x) = 1 - e^{-\lambda x}$ ($x \geq 0$).

Let $u = F_X(x) = 1 - e^{-\lambda x}$, where $U \sim \text{Uniform}(0,1)$, solve $x = -\frac{1}{\lambda} \ln(1-u)$.

Then $X = -\frac{1}{\lambda} \ln(1-U)$, since $1-U \sim U$, $X = -\frac{1}{\lambda} \ln U$.

Exercises.

Ex 1. Hint: $\chi_A(x,y) = \begin{cases} 1 & \text{if } (x,y) \in A \\ 0 & \text{otherwise} \end{cases}$

(i) since $e^{-y} > 0$, given pdf is non-negative.

To verify $P(\mathbb{R}) = 1$: $\int_{\mathbb{R}^2} P_{X,Y}(x,y) dx dy = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{-y} \chi_A(x,y) dx dy \quad (0 \leq x \leq y)$

$$= \int_0^{\infty} \left(\int_x^{\infty} e^{-y} dy \right) dx$$

$$= \int_0^{\infty} e^{-x} dx = 1.$$

(ii) $P_Y(y) = \int_{\mathbb{R}} P_{X,Y}(x,y) dx = \int_0^y e^{-y} dx = e^{-y} \int_0^y 1 dx = y e^{-y} \quad (y \geq 0)$

$$E(X|Y=y) = \int_{\mathbb{R}} x \cdot \frac{P_{X,Y}(x,y)}{P_Y(y)} dx = \int_{\mathbb{R}} x \cdot \frac{e^{-y}}{y e^{-y}} dx = \frac{1}{y} \int_0^y x dx = \frac{y}{2}, \quad E(X|Y) = \frac{Y}{2}$$

(iii) $P_X(x) = \int_{\mathbb{R}} P_{X,Y}(x,y) dy = \int_x^{\infty} e^{-y} dy = e^{-x} \quad (0 \leq x \leq y)$

$$E(Y|X=x) = \int_{\mathbb{R}} y \cdot \frac{P_{X,Y}(x,y)}{P_X(x)} dy = \int_x^{\infty} y \cdot \frac{e^{-y}}{e^{-x}} dy = x+1, \quad E(Y|X) = X+1$$

Ex 2. Hint: For $X \sim \text{Exp}(\lambda)$, $p_X(x) = \lambda e^{-\lambda x}$ ($x \geq 0$).

(i) $X \perp Y$, $p_X(x) \cdot p_Y(y) = p_{X,Y}(x,y)$. Define $A_z = \{(x,y) \in \mathbb{R}^2 \mid x+y \leq z\}$.

$$F_Z(z) = P(Z \leq z) = P(X+Y \leq z) = P((X,Y) \in A_z)$$

$$\begin{aligned} & (= \int_{-\infty}^z p_Z(u) du) & &= \iint_{A_z} p_{X,Y}(x,y) dx dy \\ & & &= \iint_{x+y \leq z} p_X(x) p_Y(y) dx dy \\ & & &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{z-y} p_X(x) dx \right) p_Y(y) dy \quad (\text{define } u=x+y) \\ & & &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^z p_X(u-y) du \right) p_Y(y) dy \\ & & &= \int_{-\infty}^z \left(\int_{-\infty}^{\infty} p_X(u-y) p_Y(y) dy \right) du \end{aligned}$$

$$\text{Therefore } p_Z(z) = \int_{\mathbb{R}} p_X(z-y) p_Y(y) dy.$$

(ii) For $X, Y \sim \text{Exp}(\lambda)$, $p_X(x) = p_Y(x) = \lambda e^{-\lambda x}$ ($x \geq 0$)

$$\begin{aligned} p_Z(z) &= \int_{-\infty}^{\infty} p_X(z-y) p_Y(y) dy \quad (z-y, y \geq 0) \\ &= \int_0^z (\lambda e^{-\lambda(z-y)}) \cdot \lambda e^{-\lambda y} dy \\ &= \lambda^2 e^{-\lambda z} \int_0^z 1 dy = \lambda^2 z e^{-\lambda z} \end{aligned}$$

(iii) For $X \sim N(\mu_X, \sigma_X^2)$, $p_X(x) = \frac{1}{\sqrt{2\pi} \sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$.

Define $Z = X+Y$ and use the formula from part (i).

Computing steps not discussed, see formal solution.

Ex 3. Hint: For a fixed value, consider rewrite into summation.

Geometric series $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $|x| < 1$.

(i) $E(X) = \sum_{n=0}^{\infty} n \cdot P(X=n)$ ($n = \sum_{k=0}^n 1$)

$$\begin{aligned} &= \sum_{n=0}^{\infty} \sum_{k=0}^n P(X=n) \\ &= 1 \cdot P(X=1) + 2 \cdot P(X=2) + 3 \cdot P(X=3) + \dots \\ &= (P(X=1) + P(X=2) + \dots) + (P(X=2) + P(X=3) + \dots) + \dots \\ &= P(X \geq 1) + P(X \geq 2) + \dots \\ &= \sum_{k=1}^{\infty} P(X \geq k). \end{aligned}$$

(ii) For $X \sim \text{Geo}(p)$, $P(X=k) = (1-p)^{k-1} \cdot p$ ($k \geq 1$).

$$\begin{aligned} P(X \geq n) &= \sum_{k=n}^{\infty} (1-p)^{k-1} \cdot p \\ &= p \cdot (1-p)^{n-1} \sum_{k=n}^{\infty} (1-p)^{k-n} \quad (\text{define } m=k-n) \\ &= p \cdot (1-p)^{n-1} \sum_{m=0}^{\infty} (1-p)^m \quad (|1-p| < 1) \\ &= (1-p)^{n-1} \end{aligned}$$

$$\begin{aligned} E(X) &= \sum_{n=1}^{\infty} P(X \geq n) \\ &= \sum_{n=1}^{\infty} (1-p)^{n-1} \quad (\text{define } m=n-1) \\ &= \sum_{m=0}^{\infty} (1-p)^m \\ &= \frac{1}{p}. \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad E(X) &= \int_0^\infty x P_X(x) dx && (x = \int_0^x 1 dy) \\
 &= \int_0^\infty \left(\int_0^x 1 dy \right) P_X(x) dx && (y \leq x) \\
 &= \iint_{y \leq x} P_X(x) dx dy \\
 &= \int_0^\infty \left(\int_y^\infty P_X(x) dx \right) dy \\
 &= \int_0^\infty P(X \geq y) dy \\
 &= \int_0^\infty 1 - F_X(y) dy
 \end{aligned}$$

For $X \sim \text{Exp}(\lambda)$, $F_X(x) = 1 - e^{-\lambda x}$:

$$\begin{aligned}
 E(X) &= \int_0^\infty (1 - (1 - e^{-\lambda x})) dx \\
 &= \int_0^\infty e^{-\lambda x} dx \\
 &= \frac{1}{\lambda}.
 \end{aligned}$$

Ex 4. Hint: Try to compute $X = F_{X|Y=y}^{-1}(U)$ for $y=0$ and $y=1$.

① $Y=0$. $F_{X|Y=0}(x) = \int_0^x (2-z) dz = 2x - x^2$.

Let $u = F_{X|Y=0}(x) = 2x - x^2$, solve $x = 1 \pm \sqrt{1-u}$.

Since $x \in [0, 1]$, $x = 1 - \sqrt{1-u}$, $X = 1 - \sqrt{1-U}$ where $U \sim \text{Uniform}(0, 1)$.

② $Y=1$. $F_{X|Y=1}(x) = \int_0^x z dz = x^2$.

Let $u = F_{X|Y=1}(x) = x^2$, solve $x = \pm \sqrt{u}$.

Since $x \in [0, 1]$, $x = \sqrt{u}$, $X = \sqrt{U}$ where $U \sim \text{Uniform}(0, 1)$.